

Building a transparent economic intelligence system for the Bay Area.

The Bay Outlook turns fragmented government releases into a reproducible county-level evidence base, an interactive dashboard, and governed economic research.

GEOGRAPHY	EVIDENCE	COVERAGE	QUALITY
Nine Bay Area counties	891 verified observations	25 cataloged indicators	83 automated tests

Problem Bay Area economic information is fragmented across agencies, geographies, methods, and release schedules. Comparing counties responsibly requires more than collecting numbers into one table.	Solution A tested system that preserves raw releases, standardizes official data, validates provenance and geography, builds reviewed comparisons, and makes freshness and evidence gaps public.
My role Defined the research framework, designed the data model, built ingestion and validation pipelines, created the analytical and editorial layers, designed the dashboard, and wrote the baseline analysis.	Result A public nine-county dashboard supported by responsible automation, immutable evidence, visible limitations, and a publication system that reserves interpretation and approval for named humans.

Python	SQLite + SQL	GitHub Actions	HTML / CSS / JS	Economic research	Data visualization
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WHAT I LEARNED

"Current" is a source-specific condition. Monthly labor data, quarterly establishment data, and annual GDP cannot be treated as one synchronous snapshot - and missing coverage is itself important information.